

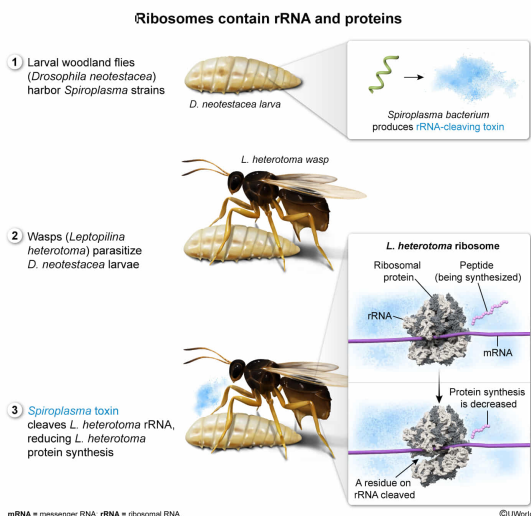
To better understand a toxin lethal to human endothelial cells, researchers are studying *Spiroplasma* bacteria known to produce a similar toxin, which confers protection against the parasitic wasp *Leptopilina heterotoma*. Larval woodland flies (*Drosophila neotestacea*) are parasitized by *L. heterotoma* and naturally harbor *Spiroplasma* strains. The *Spiroplasma* toxin cleaves a specific adenine residue in *L. heterotoma* ribosomal RNA. Given this, the toxin would be expected to have which of the following effects?

- ☒ A. Reduction in *L. heterotoma* protein synthesis [64%]
- ☐ B. Disruption of post-translational protein modification in *D. neotestacea* [11%]
- ☐ C. Introduction of beneficial mutations into *D. neotestacea* genes [3%]
- ☐ D. Interference with transcription of certain genes in *L. heterotoma* [20%]

Correct

63% answered correctly

Explanation:



Ribosomes, composed of specific proteins and **ribosomal RNA (rRNA)**, are molecular protein-synthesis machines common to **all cell types**. During **translation**, ribosomes read messenger RNA (mRNA) sequences to synthesize encoded proteins. In **eukaryotic cells** and **prokaryotic cells**, ribosomes exist in the **"free" state** in the cytoplasm. However, in eukaryotic cells only, ribosomes can exist in the **"bound" state**, where they are attached to the **rough endoplasmic reticulum**.

In this example, researchers are studying a toxin that is lethal to human **endothelial cells**. To better understand this toxin, researchers analyze *Spiroplasma* bacteria known to produce a similar toxin, which confers protection against the **parasitic** wasp *Leptopilina heterotoma*. *L. heterotoma* parasitizes the larval woodland fly *Drosophila neotestacea*, which naturally contains strains of *Spiroplasma*. The *Spiroplasma* toxin cleaves a specific **adenine** residue in *L. heterotoma* rRNA.

Therefore, given that *Spiroplasma* bacteria provide protection to the fly (*D. neotestacea*) against parasitism by the wasp (*L. heterotoma*), the cleaving of a specific adenine residue in wasp rRNA by this bacterial toxin must have negative effects on wasp ribosomes. Because ribosomes synthesize proteins, this toxin would be expected to **reduce *L. heterotoma* protein synthesis**.

(Choice B) Post-translational protein modifications are additional steps (eg, **proteolytic cleavage**) that are sometimes required before a protein becomes functional. These types of protein modifications in the fly (*D. neotestacea*) would not be affected because the toxin affects *wasp ribosomes*.

(Choice C) The question states that this toxin cleaves a specific adenine residue in wasp (*L. heterotoma*) rRNA. While production of this toxin by *Spiroplasma* bacteria benefits the fly (*D. neotestacea*) by killing the parasitic wasp, the toxin is *not* introducing **beneficial mutations** into *D. neotestacea* genes.

(Choice D) **Transcription** synthesizes mRNA from a DNA template before translation. However, this bacterial toxin cleaves rRNA comprising wasp (*L. heterotoma*) ribosomes, which are involved in translation (*not* transcription). Accordingly, this toxin interferes with translation (*not* transcription) of certain genes in the wasp.

Educational objective:

Ribosomes, composed of specific proteins and ribosomal RNA, translate messenger RNA sequences into proteins in all cells.

Time spent:0 sec

Subject:
Biology

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System:
Molecular Biology